File Edit Format Run Options Window Help

```
graph = {
    'A': ['B', 'C'],
    'B': ['D', 'E'],
    'C': ['F', 'G'],
    'D': [],
    'E': [],
    'F': [],
    'G': []
}

def dfs(node, goal, visited, path):
    visited.add(node)
    path.append(node)

    print("Visiting Node:", node)

    if node == goal:
        print("Goal Reached!")
        return True

    for neighbor in graph[node]:
        if neighbor not in visited:
            if dfs(neighbor, goal, visited, path):
                return True

    path.pop()
    print("Backtracking from:", node)
    return False

visited = set()
path = []

print("DEPTH FIRST SEARCH")
print("Starting Node: A")
print("Goal Node: G")

if dfs('A', 'G', visited, path):
    print("Path Found:", " -> ".join(path))
    print("Total Steps:", len(path) - 1)
else:
    print("Goal not found")
```

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File Edit Shell Debug Options Window Help

```
Python 3.13.1 (tags/v3.13.1:0671451, Dec  3 2024, 19:06:28) [MSC v.1942
64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
========== RESTART: C:/Users/user/Downloads/EX-AI-1.py ==========
========
DEPTH FIRST SEARCH
Starting Node: A
Goal Node: G
Visiting Node: A
Visiting Node: B
Visiting Node: D
Backtracking from: D
Visiting Node: E
Backtracking from: E
Backtracking from: B
Visiting Node: C
Visiting Node: F
Backtracking from: F
Visiting Node: G
Goal Reached!
Path Found: A -> C -> G
Total Steps: 2
Visited Nodes: {'D', 'G', 'A', 'B', 'E', 'C', 'F'}
>>>
```

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